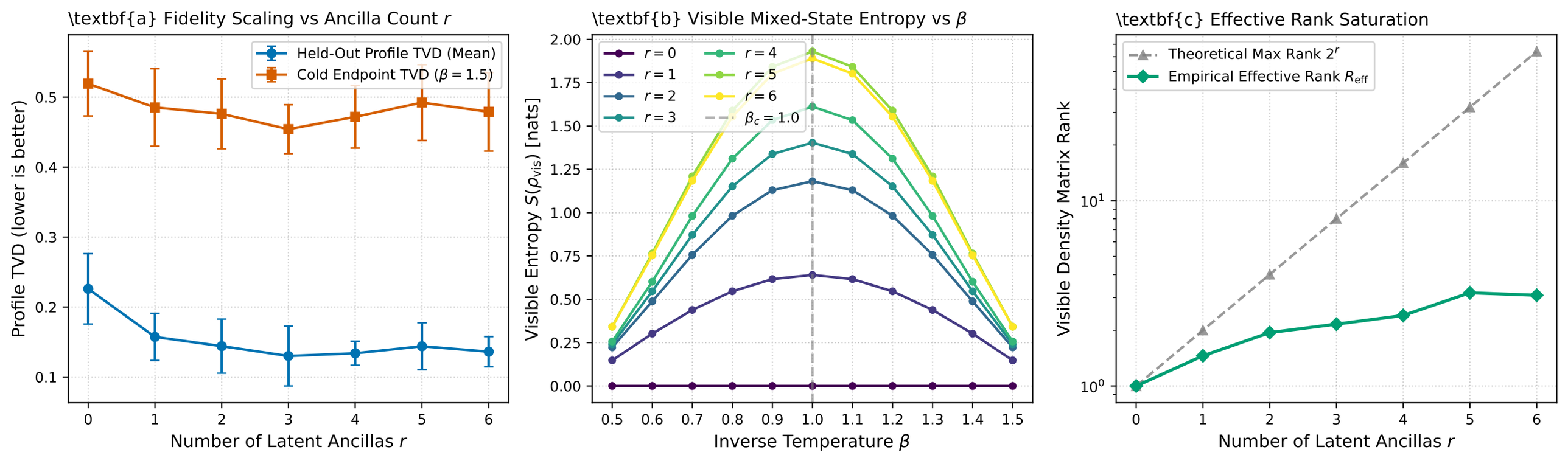
\textbf{a} Fidelity Scaling vs Ancilla Count $r$
Held-Out Profile TVD (Mean)
Cold Endpoint TVD ($\beta = 1.5$)
Profile TVD (lower is better)
Number of Latent Ancillas $r$
\textbf{b} Visible Mixed-State Entropy vs $\beta$
$r = 0$
$r = 1$
$r = 2$
$r = 3$
$r = 4$
$r = 5$
$r = 6$
$\beta_c = 1.0$
Visible Entropy $S(\rho_{\text{vis}})$ [nats]
Inverse Temperature $\beta$
\textbf{c} Effective Rank Saturation
Theoretical Max Rank $2^r$
Empirical Effective Rank $R_{\text{eff}}$
Visible Density Matrix Rank
Number of Latent Ancillas $r$